Earth as a Field of Admissible Futures

Geoengineering, Biomass, and Civic Simulation

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Introduction: The Planet Has No Single Set Point

Geoengineering is usually imagined through a single image: a machine, or a coordinated set of machines, acting on a single planetary variable to move it toward a preferred value. The variable is almost always temperature, whether the mechanism is stratospheric aerosol injection, solar radiation management, or a nuclear-powered installation of the kind Neal Stephenson imagines in *Termination Shock*. In that novel, a private actor builds an intervention large enough to alter the climate unilaterally, forcing every affected government, institution, and population to respond to a fact rather than deliberate over a proposal. The novel is not a source of evidence for anything argued here, but it dramatizes something true about the ordinary shape of the geoengineering imagination: one apparatus, one variable, one planet moved toward one target.

This book replaces that image with a different one. It treats geoengineering as a portfolio of interventions distributed across water, ice, clouds, ocean currents, biomass, nutrients, and infrastructure, each acting on a different part of the Earth system, each subject to its own constraints and failure modes, and none of them reducible to a single planetary thermostat. Chapter 1 has already shown why the thermostat model fails even on its own terms: there is no established optimal global temperature, because the systems that might be optimized (biomass, biodiversity, agricultural productivity, coastal stability, ocean circulation) respond to temperature differently and do not share a maximum. Chapter 2 has shown why the most obvious fallback objective, maximizing total biomass, fails in the same way, since biomass existing in the wrong place or beyond a receiving system's capacity to incorporate it can represent damage rather than health regardless of its quantity.

Given those two results, the central question this book asks is not "what temperature should Earth be" but "which climatic and ecological futures should remain reachable, for whom, under what risks, and with what capacity for correction." This question does not collapse into a scalar, and it is not meant to. It is meant to keep visible a plurality of outcomes, populations, and ecosystems that a single target necessarily suppresses, and to hold every intervention this book considers, including the ones it proposes, to the standard of preserving that plurality rather than collapsing it toward one preferred future.

The book's remaining parts follow the shape of that question rather than the shape of a single mechanism. Part I, of which this introduction and the following four chapters are the opening movement, establishes the conceptual ground: the fiction of the optimal temperature, the distinction between biomass and biospheric health, climate change as a mismatch between committed systems and revised regularities, and the vocabulary of continuation geometry (reachability, viability, recoverability, and path dependence) that the rest of the book uses to evaluate specific proposals. Part II turns from concept to method, asking how competing interventions, models, and preferences can be explored and compared before any of them is committed to at planetary scale; it does this by adapting two existing frameworks, *Simulation as Civic Infrastructure* and *Civilizations as Hypotheses*, to the specific problem of planetary intervention, rather than developing a new theory of governance from scratch. Parts III through VI then work through specific interventions in turn: active polar albedo maintenance, oceanic continuation engineering, constructed terrestrial productivity, and planetary material circulation, each held to the standards Chapters 1 and 2 establish rather than judged by whether it lowers a single number. Part VII closes with governance, failure, and repair, asking not only who is authorized to alter a shared atmosphere but how interventions can be designed from the outset to be terminated, corrected, or reversed without producing a second crisis in the act of undoing the first.

Stephenson's Schmidt gets one thing right that this book preserves: once an intervention becomes materially credible, it forces positions that abstract deliberation could defer indefinitely. This book's proposals aim for the same forcing function without the same unilateralism, by using shared simulation, explicit governance structure, and termination planning to make positions visible and contestable before, rather than after, an intervention becomes irreversible.

Part I

Beyond the Planetary Thermostat

Chapter 1

The Fiction of the Optimal Temperature

Climate policy is conventionally organized around a single number: the increase in global mean surface temperature relative to a preindustrial baseline. Targets are expressed in fractions of a degree. Success or failure is reported as a threshold crossed or held. This framing has been indispensable for coordinating international attention, and it will remain indispensable as a diagnostic variable throughout this book. But a diagnostic variable is not the same thing as an engineering objective, and the temptation to treat it as one has quietly organized a great deal of geoengineering discussion around a question that has no defensible answer: what temperature should Earth be?

No such temperature has been established, and there is good reason to think none exists in the form the question presupposes. The difficulty is not that the answer is currently unknown but likely to be found with better models. The difficulty is that "optimal" requires an objective, and the objectives that might be optimized are neither singular nor aligned with one another.

Consider a partial list of the quantities a planetary temperature affects: total biomass, biodiversity, agricultural productivity, human heat tolerance, disease range, coastal land area, ocean primary production, ice sheet stability, freshwater availability, and the frequency of extreme weather. Each of these responds to temperature differently, and not always monotonically. A given warming increment may lengthen a growing season in one latitude while shortening it in another. It may increase net photosynthesis in a boreal forest while triggering dieback in a tropical one. It may improve conditions for one commercially important fish species while degrading them for another sharing the same water column. There is no reason to expect these responses to share a maximum, and considerable reason to expect they do not.

This is not a claim that all temperatures are equally good, or that climate change is not a serious problem. It is a claim about the shape of the problem. The question "is this level of warming dangerous" can be answered, often clearly, once a specific system and a specific form of danger are named: coastal infrastructure committed to a stable sea level, agricultural regions committed to a rainfall pattern, coral reefs committed to a narrow thermal tolerance. The question "what is the optimal global temperature" cannot be answered in the same way, because it asks for a single scalar to summarize outcomes that are irreducibly plural and frequently in tension.

A particularly instructive case is the relationship between temperature and biomass, because it is often assumed, implicitly, that warming is either straightforwardly bad for the biosphere or straightforwardly good for plant growth, and neither assumption survives close examination. Some of the most biologically productive regions in the present ocean are cold, not because coldness directly produces life, but because low temperatures coincide with circulation regimes that deliver nutrients to the surface. Winds and ocean circulation bring cold, nutrient-rich water upward into sunlit layers where phytoplankton can use it. This upwelling, rather than coldness by itself, makes the Southern Ocean and the eastern boundary currents of the Pacific exceptionally productive marine environments. A warmer ocean can therefore hold more thermal energy while supporting less primary production in exactly the regions where upwelling currently sustains large food webs.

This does not establish that a colder Earth would be biologically superior to a warmer one. It establishes that the relationship between temperature and productivity runs through intermediate variables, chiefly wind-driven circulation and nutrient delivery, that do not scale simply with temperature in either direction. Warming in one region can open new nutrient pathways even as it closes others elsewhere. Both very warm and very cold planetary states have supported extensive marine productivity under the right circulation conditions, and neither state guarantees it.

The same caution applies to land. Extended growing seasons and elevated atmospheric carbon dioxide can increase certain crop yields in specific regions, holding water and nutrients constant, which they typically are not. Warming also raises evapotranspiration, shifts rainfall patterns, expands the range of some agricultural pests and pathogens, and pushes many staple crops toward or past their thermal tolerance during critical growth stages. Whether a given region experiences net agricultural benefit or net agricultural loss from a given warming increment depends on its starting climate, its crops, its water infrastructure, and the rate at which the change occurs relative to the capacity of farmers, seed varieties, and irrigation systems to adapt. There is no global answer, only a distribution of regional ones, and averaging that distribution would conceal precisely the differences an intervention must preserve.

If temperature does not determine a single correct value for biomass, biodiversity, or productivity, it follows that biomass itself cannot serve as a simpler proxy objective in its place. This might seem to invite a fallback position: perhaps the goal should simply be to maximize the total quantity of living matter the planet supports, since more biomass would seem to indicate a more vigorous biosphere. Chapter 2 develops in detail why this fallback fails, using two paired cases, the early closure of the Antarctic krill fishery and the record-breaking growth of the Great Atlantic Sargassum Belt, to show that biomass existing in the wrong place, at the wrong concentration, or beyond a receiving system's capacity to incorporate it can represent ecological damage rather than ecological health, regardless of the total tonnage involved. The short version of that argument, needed here only to close off the fallback, is that quantity and function are different measurements. A planet can gain biomass while losing the ecological relationships that made a smaller quantity of it valuable.

This has direct consequences for how climate risk should be framed. It is common, and not wrong, to note that a large share of visible climate damage concentrates on low-lying coastal populations facing sea-level rise. But treating coastal flooding as the central problem risks understating everything else at stake, and it also risks implying that a strategy calibrated to protect coastlines has thereby addressed climate change. A more general and more accurate framing is that climate risk arises wherever human or ecological systems have become committed, often over decades or centuries, to a set of climatic regularities that a changing climate is now revising. Coastal cities are one instance of this commitment, made visible by sea walls, property values, and evacuation maps. Agricultural regions committed to a rainfall pattern are another, less visible until a multi-year drought exposes it. Coral reefs committed to a narrow thermal tolerance are a third, revealed by bleaching events rather than infrastructure failure. Fisheries committed to a seasonal ice cycle, water utilities committed to a snowpack-fed river, and insurance markets committed to historical loss tables are further instances of the same underlying structure. Chapter 3 develops this commitment-mismatch framing directly; it is introduced here only to show why coastal flooding, however serious, is a special case of a general problem rather than the problem itself.

None of this implies that a single global temperature target is useless as a coordinating device. Diagnostic variables earn their keep by being measurable, comparable across regions and years, and correlated with a wide range of underlying risks, and global mean temperature satisfies all three conditions well enough to remain central to monitoring and to international agreement. The error is not in measuring it. The error is in treating its reduction as identical to the achievement of planetary health, and in treating any proposed intervention primarily by whether it moves that one number in the preferred direction.

The chapters that follow adopt a different governing question. Rather than asking what temperature Earth should have, they ask which climatic and ecological trajectories should remain reachable, for which populations and ecosystems, under what risks, and with what capacity for correction if the intervention turns out to be mistaken. This question does not resolve into a single target, and it is not meant to. It is meant to keep visible the plurality of outcomes that a single scalar necessarily suppresses, and to hold every intervention considered in this book, including the ones proposed here, to the standard of preserving that plurality rather than collapsing it toward one preferred future.

Chapter 2

Biomass Is Not Biospheric Health

In 2025, the Antarctic krill fishery reached its annual 620,000-tonne trigger level early for the first time. It did so again in 2026. At nearly the same time, the tropical Atlantic was producing Sargassum on an extraordinary scale. The Great Atlantic Sargassum Belt reached a record abundance in 2025 and remained near that level in 2026, when the Caribbean Sea and Gulf of Mexico experienced record regional accumulations. Read only as measurements of biomass, these developments might appear to point in opposite directions. One concerns anxiety over the removal of marine organisms; the other concerns an apparently vigorous expansion of marine life.

Ecologically, however, they describe the same structural problem.

Antarctic krill remain abundant when measured across the immense area of the Southern Ocean. The 620,000-tonne fishery trigger represents a small fraction of estimated regional biomass. That aggregate comparison has consequently become central to arguments that the fishery remains sustainable. Yet whales, penguins, seals, fish, and seabirds do not consume a regional biomass estimate. They consume krill that must be present within reachable foraging distance, at the correct depth, during the correct season, and in sufficient density to repay the energy expended finding and capturing them.

A penguin feeding chicks cannot substitute krill located hundreds of kilometres away for krill removed near its colony. A recovering whale population cannot consume the portion of the statistical stock that is inaccessible during migration. Biomass may exist at the scale of the management model while failing to exist operationally at the scale of the animal.

The recent fishery closures sharpen this distinction because they followed a change in spatial governance. Until the end of the 2024 season, a conservation measure distributed the Area 48 catch among several subareas. When that measure lapsed, the aggregate trigger level remained, but the restrictions governing where the catch could be taken disappeared. The numerical ceiling did not rise. The geometry beneath it changed.

Fishing vessels could concentrate more effectively in krill-rich locations. Those locations are rich for the same reason that predators gather there: currents, bathymetry, seasonal production, and krill behaviour produce dense, accessible swarms. Removing spatial restrictions therefore increased the fleet's effective capability without increasing its nominal authorization. A limit designed at the scale of the fishery could remain intact while protection at the scale of the food web deteriorated.

The early closures do not, by themselves, prove that Antarctic krill have been depleted below a biologically sustainable global or regional level. They establish something more precise: industrial harvesting capacity can now traverse the permitted catch space faster and concentrate within it more strongly. Whether that capability produces ecosystem damage depends on where and when extraction occurs, which predators share those locations, how rapidly krill are replenished, and whether climate-driven changes in sea ice and circulation are already narrowing the predators' alternatives.

This is an ecological admissibility problem. Krill can exist, be scientifically observed, enter a biomass estimate, and remain legally harvestable without remaining available to every organism whose continuation depends on them. Existence does not imply reachability. Reachability does not imply availability at the necessary time. Availability does not imply that extraction is recoverable. An aggregate stock can remain above a precautionary threshold while local food-web continuations become progressively harder to sustain.

Sargassum reveals the inverse failure.

Floating Sargassum is not inherently pollution. In the open ocean it forms a mobile habitat supporting fish, crabs, molluscs, microorganisms, seabirds, and sea turtles. It captures carbon through photosynthesis and creates shelter and feeding surfaces in waters that might otherwise possess relatively little structural complexity. Within the Sargasso Sea, its presence is part of an established ecological regime.

The Great Atlantic Sargassum Belt is a different spatial phenomenon. Since 2011, enormous seasonal aggregations have repeatedly formed across the tropical Atlantic, extending from West Africa toward the Caribbean and Gulf of Mexico. The causes remain under investigation and likely include changing circulation, winds, rainfall, nutrient delivery, river discharge, atmospheric deposition, upwelling, and warmer waters. No single cause adequately explains every year or region.

When Sargassum reaches the coast in extreme quantities, its ecological function can reverse. Dense mats reduce light reaching corals and seagrasses, obstruct the movement of animals, alter water circulation, and physically smother shallow habitats. Once stranded or trapped near shore, the accumulated material begins to decompose. Microbial decomposition consumes dissolved oxygen, while the decaying biomass releases hydrogen sulfide, ammonia, and other compounds. Material that served as habitat offshore becomes an anoxic and sometimes toxic burden at the coastline.

Nothing has disappeared during this transition. The biomass remains present, indeed, its excessive presence is the problem. What changes is the receiving system's capacity to incorporate it.

A beach, mangrove margin, seagrass bed, municipal collection service, or coastal decomposition zone can process some arrival of organic matter. Beyond a certain rate, however, incoming biomass exceeds the available oxygen, space, labour, transport, storage, and decomposition pathways. The surplus does not become additional ecological wealth merely because it contains carbon and nutrients. It becomes maladmitted biomass: biological material arriving in a quantity, location, or state for which the receiving system has no viable continuation.

The krill and Sargassum cases therefore cannot be understood by placing their tonnage into a common planetary account. One system may retain a large total stock while losing strategically located biomass. Another may gain tens of millions of tonnes while becoming less habitable where that biomass accumulates. A global calculation could record the continued existence of Antarctic krill and the proliferation of Atlantic Sargassum as evidence that marine biomass remains plentiful. Such a calculation would miss the actual failures occurring in both systems.

The distinction is not between "too little life" and "too much life." It is between biomass as quantity and biomass as ecological capability.

Biomass becomes capability only through relations. It must be reachable by the organisms that use it, synchronized with their life cycles, compatible with the chemistry of the receiving environment, and supported by pathways that redistribute its matter after death. Its value depends on what can eat it, inhabit it, transform it, transport it, decompose it, or preserve it. The same tonne of biological material can function as food, shelter, carbon reservoir, industrial feedstock, oxygen-consuming waste, or toxic obstruction depending on its position within those relations.

This is why the search for an "optimal biomass level for Earth" cannot be reduced to maximizing a global total. Even if total planetary biomass could be estimated reliably, the number would conceal its composition and topology. A biosphere dominated by a small number of rapidly proliferating organisms might contain substantial biomass while possessing reduced biodiversity and resilience. A region of exceptional primary production might export its carbon efficiently into deep water, support a complex food web, accumulate as undecomposed waste, or collapse into hypoxia. Equal quantities do not imply equal ecological function.

Temperature enters this problem as one condition among many. A warmer Earth may increase growing seasons and biological production in some regions. It may also strengthen ocean stratification, reducing the movement of nutrients from deeper water into the sunlit surface. Some of the most productive marine environments are cold not because coldness alone creates biomass, but because cold-water circulation and upwelling maintain access to nutrients. Conversely, warm nutrient-rich waters can support extraordinary blooms whose eventual decomposition damages the systems around them.

The relevant question is therefore not whether warmth or cold produces more life in the abstract. It is which combinations of temperature, nutrients, circulation, oxygen, acidity, seasonality, and food-web structure preserve productive biomass as an available and renewable ecological relation.

This distinction also changes how geoengineering should be evaluated. An intervention that increases net primary production cannot be presumed beneficial without asking what grows, where it grows, which organisms can use it, and what happens when it dies. An intervention that reduces biomass in one region cannot be presumed harmful without asking whether it suppresses a pathological bloom or prevents an anoxic transition. Gross production, carbon fixation, standing stock, biodiversity, harvestable yield, and ecological continuity are different quantities. They may sometimes rise together, but no general rule requires them to do so.

The Caldera Reactor and its associated biomass systems begin from this problem of incorporation. Managed kelp farms would provide a relatively predictable marine feedstock grown within an infrastructure designed to receive and process it. Excess Sargassum would enter as a secondary, disturbance-responsive stream. The purpose would not be to reward or perpetuate anomalous blooms. It would be to intercept some fraction of the biomass before coastal deposition converts it from offshore habitat into an oxygen-consuming burden.

The distinction between managed kelp and opportunistic Sargassum is essential. If the reactor became economically dependent on pathological Sargassum abundance, it could create an institutional interest in the continuation of the ecological failure it was meant to relieve. Surrounding kelp farms instead provide a controllable baseline substrate, while the adaptive processing system offers additional capacity when anomalous biomass arrives. The infrastructure remains viable without requiring the disturbance.

Processing alone does not guarantee ecological benefit. Collection vessels expend energy. Wet biomass is expensive to transport and dewater. Sargassum may contain salt, sand, plastics, arsenic, heavy metals, or decomposed material that complicates conversion. Biocrude returns carbon to the atmosphere when burned, whereas durable bioplastics or stable carbon products may retain it longer. The relevant carbon balance depends on the complete pathway and on what would have happened to the Sargassum without interception.

Nevertheless, the conceptual role of the reactor is clear. It supplies an alternative continuation for biomass that a coastal system cannot incorporate. It changes the material's admissible future: from uncontrolled deposition and decomposition toward differentiated products, recoverable energy, durable carbon forms, or carefully managed residues. Its success would be measured not by the gross amount of biomass consumed but by the ecological damage avoided, useful displacement achieved, and carbon retained.

Krill management poses the complementary task. There the objective is not to create an alternative use for surplus biomass, but to preserve local availability within an existing food web. Spatial catch constraints may appear economically inefficient because they prevent vessels from concentrating where harvesting is easiest. Ecologically, that inefficiency is precisely their function. By dispersing extraction, they preserve distinctions among feeding regions and reduce the probability that industrial capability will collapse a locally essential swarm into an aggregate statistic.

The two cases reveal a general principle:

> *Biospheric health depends not on how much living material exists, but on whether biomass remains appropriately distributed, reachable, transformable, and incorporable across the systems whose continuation depends upon it.*

This principle will govern the interventions considered throughout this book. Polar ice will not be evaluated merely by tonnes frozen, but by reflective area, seasonal persistence, ecological function, and the larger ice regimes thereby preserved. Rainforest generators will not be evaluated merely by vertical biomass, but by habitat complexity, nutrient continuity, maintenance burden, and increasing ecological autonomy. Kelp cultivation will not be evaluated solely by growth rate, but by its effects on marine food webs, carbon pathways, oxygen, harvest cycles, and material use. Geoengineering will not be treated as the production of desirable quantities in isolation. It will be treated as the design of relations through which those quantities acquire, or lose, the capacity to sustain a world.

Chapter 3

Climate Change as Commitment Mismatch

Sea-level rise draws a disproportionate share of public and political attention among the consequences of climate change, and for good reason. It is visible, cumulative, and largely irreversible on human timescales, and it threatens some of the most densely populated and economically valuable land on Earth. But the reason sea-level rise feels distinctively urgent is not that it is a separate category of harm from everything else climate change produces. It is that it makes unusually legible a structural problem that runs through nearly every consequence of a changing climate: human and ecological systems become committed to a set of climatic regularities, often over decades or centuries, and climate change revises those regularities after the commitment has become expensive or impossible to undo.

Commitment, in this sense, is not a choice made carelessly. It is the ordinary and often rational outcome of building anything durable. A coastal city is not built in a single year by people ignorant of sea level; it accumulates across generations, each addition assuming that the shoreline, tidal range, and storm-surge frequency observed at the time of construction will remain approximately stable. A seawall calibrated to a particular flood return period, a subway system built below a particular water table, a port whose depth and layout assume a particular sea level, and the property values, tax base, and insurance markets that depend on all three, together constitute a vast, distributed bet that the coastline of the recent past will resemble the coastline of the future. That bet was reasonable for most of the period during which it was made. It has become progressively less reasonable without any single decision-maker having chosen to make it so.

The same structure appears wherever commitment and climatic regularity diverge, whether or not the result resembles a coastline.

Agricultural regions commit to a rainfall pattern through crop selection, irrigation infrastructure, land tenure, and the accumulated knowledge of farmers about when to plant and what to expect. A region that has reliably received a wet season followed by a dry season will develop crop varieties, storage systems, and credit cycles suited to that pattern, and those investments are not easily redirected when the pattern shifts. A multi-year drought does not simply reduce yield in the years it occurs; it can exhaust the assumptions embedded in seed stock, debt schedules, and water rights that were calibrated to a climate the region no longer has.

Water utilities in mountainous and high-latitude regions commit to a snowpack that accumulates in winter and releases meltwater gradually through spring and summer, smoothing supply across a season that would otherwise alternate between flood and drought. Reservoir capacity, treatment infrastructure, and downstream agreements among competing users are sized to this seasonal release.

When warming shifts precipitation from snow to rain, or accelerates snowmelt into a shorter and earlier pulse, the physical volume of water available across a year may change relatively little while its timing changes enough to overwhelm infrastructure built for a different release curve.

Coral reefs commit, in an ecological rather than an institutional sense, to a narrow band of sea-surface temperature within which the symbiotic algae that reefs depend on for energy and color can survive. This commitment is written into the reef's own biology rather than into concrete or contracts, but it produces the same structural vulnerability: a system optimized for a historical regularity has no graceful response available when that regularity shifts faster than the system's own capacity for adaptation or migration.

Insurance and reinsurance markets commit to actuarial tables built from historical loss data, on the assumption that the frequency and severity of future events will resemble the frequency and severity of past ones closely enough for the tables to remain solvent. Fisheries commit to seasonal ice cover, migration timing, or spawning-ground temperature. Disease vectors commit, and human populations commit around them, to a range within which a given mosquito, tick, or pathogen can or cannot survive a given latitude's winter. In each case, a system, whether institutional, ecological, or actuarial, has organized itself around a regularity that persisted long enough to be treated as a stable feature of the world, and climate change is now revising that regularity on a timescale shorter than the system's ability to reorganize around a new one.

This framing has several consequences for how climate risk should be assessed and how interventions should be evaluated.

First, it explains why the same magnitude of physical change can produce wildly different amounts of harm depending on what has been built around the prior condition. A degree of warming that a sparsely settled, agriculturally flexible region absorbs with modest adjustment can devastate a densely built, heavily indebted, institutionally rigid region experiencing the identical physical change. Vulnerability is not a direct function of exposure to a hazard; it is a function of exposure combined with the depth and inflexibility of prior commitment. This is why coastal megacities, snowpack-dependent water systems, and narrow-tolerance ecosystems recur as the most frequently cited examples of climate risk: not because their physical exposure is uniquely severe in an abstract sense, but because their commitments are unusually deep, unusually expensive to relocate, and unusually slow to renegotiate.

Second, it clarifies why adaptation is not simply a matter of technical capacity but of timing. A commitment can in principle be revised, a seawall raised, a crop variety changed, a reservoir rebuilt, a reinsurance table repriced, but revision takes time, capital, and institutional coordination that themselves have a rate. If the underlying climatic regularity changes faster than the commitment can be revised, the system experiences the change as a shock even though the physical trend that produced it may have been gradual and, in principle, foreseeable for decades. Much of what appears as sudden climate catastrophe is better described as the delayed collision between a slow physical trend and a commitment whose revision rate was slower still.

Third, and most directly relevant to the chapters that follow, this framing reorients what a geoengineering intervention should be evaluated against. An intervention should not be judged primarily by whether it returns some global variable toward a historical value. It should be judged by whether it preserves, extends, or safely revises the reachability of the specific commitments that populations and ecosystems have already made, and by whether it does so without creating new commitments whose own revision, should the intervention need to be altered or withdrawn, would prove equally rigid. An intervention that stabilizes a coastline while making the underlying climatic trend worse elsewhere has not resolved a commitment mismatch; it has relocated one. An intervention that succeeds only so long as it continues indefinitely has not restored a lost regularity; it has substituted a new commitment, to the intervention itself, for the old one, and that substitution carries its own risks if the intervention is later withdrawn, defunded, or rendered obsolete.

This last point anticipates the central concept of the following chapter. An intervention that a system comes to depend on cannot simply be removed once it has been in place long enough for other systems to reorganize around it. Stopping it does not restore the world that existed before it began; it produces a new transformation, potentially more disruptive than either the original problem or the intervention's continued operation. This is termination shock in its general form, and it is one of the central risks that continuation geometry, developed next, is built to make explicit and evaluable before any intervention proposed in this book is undertaken at scale.

Chapter 4

Continuation Geometry

The preceding chapters have used several words informally that now need to be given exact meaning: reachable, viable, recoverable, committed. Chapter 3 leaned on all four without defining any of them, because the argument at that stage needed only their intuitive sense. Formalizing them here is not a detour into abstraction for its own sake. It is what allows every intervention discussed in the rest of this book, from a nuclear-powered ice system to a routed iceberg to a planetary infrastructure network, to be evaluated against a single, consistent standard rather than assessed impressionistically case by case.

The central idea is that an intervention does not simply move a system from one state to another. It changes which later states remain available to reach. A climate intervention, a piece of infrastructure, or an ecological disturbance can be described not only by where it puts a system today, but by the shape of the space of futures that system can subsequently move through. This chapter calls that space, and its evolution under intervention, continuation geometry.

Reachability

A state is reachable from a given starting point if some sequence of physically and institutionally permitted transitions connects the two. Reachability is the weakest of the concepts developed in this chapter, and it is important to establish it first precisely because it is weak: many proposals are defended by showing that a desired outcome is reachable in principle, without showing that it is reachable under the constraints that actually govern the system, or that reaching it would leave the system able to do anything else afterward.

Consider polar sea ice. It is reachable, in the narrow sense, to freeze a given volume of seawater given sufficient energy and time; nothing in the physics forbids it. But this observation alone says nothing about whether the resulting ice performs any useful climatic function, whether the energy expenditure and waste heat involved are compatible with the surrounding system's other commitments, or whether the frozen state persists long enough to matter. Reachability answers the question "can this state be reached at all." It does not answer "should it be," "can it be reached without foreclosing other states," or "can it be maintained once reached." Those questions require the stronger concepts that follow.

Viability

A state, or more precisely a trajectory through a sequence of states, is viable if the system can occupy it while continuing to satisfy the constraints that define its ongoing function. A power grid is not viable merely because it can be made to deliver electricity for a single afternoon; it is viable if it can deliver electricity continuously while maintaining frequency, voltage, and reserve margins within the tolerances its connected systems require. A coral reef is not viable merely because individual corals can survive a single marine heatwave; it is viable if the reef's net calcification, recruitment, and recovery processes can keep pace with disturbance over the timescales on which the reef must persist.

Viability is therefore a claim about a region of state space, not a single point within it. Engineers and ecologists both use versions of this concept, often called an admissible region or a safe operating envelope, to describe the set of conditions under which a system continues to function rather than degrading, collapsing, or requiring outside rescue. An intervention can move a system into a reachable state that lies outside its viable region: the state exists, the system can be pushed into it, but the system cannot remain there and continue doing what it needs to do. This is the failure mode that a purely reachability-based evaluation misses and that a viability-based evaluation is designed to catch.

Viability regions are not fixed. They shift as a system's surrounding conditions change, and this is precisely what climate change does to the systems described in Chapter 3. A coastal city's viable region, defined by the combination of sea level, storm frequency, and infrastructure capacity within which it can continue to function without catastrophic loss, contracts as sea level rises, even if no single event pushes the city outside it. The commitment mismatch discussed in the previous chapter can now be restated more precisely: a system's commitments were made on the assumption of a particular viable region, and climate change is shrinking or relocating that region faster than the system can revise its commitments to match.

Recoverability

A state is recoverable if, once a system enters it, some available and sufficiently fast process can return the system to its viable region before irreversible loss occurs. Recoverability is what separates a viability excursion, a temporary departure from the admissible region that resolves before permanent damage accrues, from a genuine collapse.

Recoverability depends on rate as much as on mechanism. A drought that pushes a rain-fed agricultural region outside its viable moisture range is recoverable if the region has water reserves, alternative crops, or credit sufficient to bridge the gap until rainfall resumes. The same drought, extended long enough, exhausts those reserves and crosses into unrecoverable territory: soil degrades past the point of restoration within a farming generation, seed stock is consumed rather than replanted, debt forecloses land that would otherwise have supported recovery. The physical hazard did not change in kind between the recoverable and unrecoverable cases; what changed was whether a sufficiently fast countervailing process remained available.

This gives recoverability a strict boundary rather than a fuzzy one, at least in principle: for a given system, there exists some maximum duration and severity of excursion beyond which no available process returns it to viability in time. Locating that boundary precisely is often difficult, since it depends on reserves, institutional capacity, and processes that are only partially observable in advance. But the existence of the boundary, and the fact that it can in principle be estimated and monitored, is what allows recoverability to function as an engineering standard rather than merely a retrospective judgment applied after a collapse has already occurred.

Path Dependence

The three concepts above describe a system's relationship to a given state or trajectory at a given time. Path dependence describes how that relationship changes as a consequence of the route by which the system arrived. Two systems occupying what appears to be the identical present state can have different viable regions and different recoverability boundaries going forward, because the route each took to reach that state altered capacities, dependencies, or reserves that are not visible in the state description alone.

A reservoir at fifty percent capacity that arrived there through a single dry season behaves differently, going forward, than a reservoir at fifty percent capacity that arrived there through a decade of gradually declining snowpack, because the second reservoir's watershed, downstream ecosystems, and water-rights agreements have already begun reorganizing around a lower baseline in ways the first reservoir's have not. The present state is the same number; the continuation geometry is not.

Path dependence is what makes intervention timing, not only intervention magnitude, a first-order variable throughout this book. An intervention introduced early, while a system's viable region is still close to its historical range, can often be modest in scale and still effective. The identical intervention introduced later, after the system's surrounding dependencies have reorganized around a degraded baseline, may need to be far larger to achieve the same effect, or may no longer be sufficient at any scale, because the system's recoverability boundary has itself moved.

Termination Shock as the Paradigmatic Case

These four concepts converge most sharply in the phenomenon that gives Neal Stephenson's novel its title, and that this book has referenced only informally until now. An intervention is introduced into a system. The system's surrounding dependencies, infrastructure, agriculture, institutions, and in some cases entire ecosystems, begin adapting to the intervention's presence, not to the prior natural state it was designed to approximate. This adaptation is itself a form of path dependence: the system's continuation geometry reorganizes around the intervention as a standing feature of its environment rather than as a temporary correction.

Once this reorganization has occurred, removing the intervention does not restore the system to the state that existed before the intervention began. It produces a new transition, from a state adapted to the intervention's presence to a state that must suddenly readapt to its absence, and this transition can be more severe, and can cross recoverability boundaries the original problem never approached, than either the initial climatic disturbance or the intervention's continued operation would have.

This is why termination shock cannot be treated as an unfortunate side effect to be managed after the fact. It is a predictable consequence of path dependence applied to any intervention sustained long enough for dependent systems to reorganize around it, and it means that every intervention discussed in this book must be evaluated not only by its effect while operating, but by the reachability, viability, and recoverability of the transition that would follow its slowing, failure, or deliberate withdrawal. An intervention whose benefits depend on indefinite continuation has not solved a commitment mismatch. It has substituted a new commitment, to itself, for the one it was meant to relieve, and that substitution must be priced into the intervention's design from the outset rather than discovered during an eventual termination.

Continuation Geometry as an Evaluative Standard

Taken together, reachability, viability, recoverability, and path dependence supply the standard the rest of this book applies. A proposed intervention will be asked: what states does it make reachable that were not reachable before, and at what cost to states that were reachable and now are not. What is its effect on the viable region of the systems it touches, including systems it was not explicitly designed to affect. What is the recoverability boundary of the trajectories it creates, and how does that boundary compare to the boundary of the trajectory it replaces. And what does the intervention's own termination look like, given the path-dependent reorganization its continued operation would produce.

This standard does not by itself tell us which interventions to pursue. It tells us what a defensible answer to that question must account for, and it rules out the two simplest but inadequate answers this book has already rejected: that an intervention is justified because it moves a single planetary variable in a preferred direction, and that an intervention is justified because it increases some aggregate quantity, such as biomass, without regard to the geometry of the commitments built around it. Part II now turns to the practical problem this standard creates: how competing interventions, each with its own effect on reachability, viability, recoverability, and path dependence, can be explored, compared, and contested before any of them is committed to at planetary scale.

Part I Status

Part I is complete: Introduction and Chapters 1–4. Parts II–VII (Chapters 5–27) and the Conclusion remain to be drafted.